

Crown Stores

Retail Management System

System Guide & Presentation Document

CSRMS Version 2.0

This document explains how the Crown Stores Retail Management System works — for store owners, managers, and stakeholders. It covers business benefits, daily workflows, user roles, and a clear overview of how the technology supports your operations.

Prepared for

Crown Stores — Directors, Branch Managers, and Operations Team

Document purpose

- Explain what the system does and why it matters for the business
- Show how each staff role uses the system day to day
- Describe the complete flow from stocking shelves to closing the till
- Provide a simple technical overview for decision-makers

1. Executive Summary

Crown Stores CSRMS v2.0 | Page 1

Crown Stores Retail Management System (CSRMS) is a secure, web-based platform that replaces manual record-keeping with a single digital system. Staff access it through a web browser — no special software installation is needed beyond a standard computer or tablet connected to your store network.

The system gives you real-time visibility into stock levels, sales performance, procurement history, and cashier accountability. Every important action is recorded automatically, reducing errors and improving trust across the business.

Key business benefits

- Real-time inventory — always know what is in stock and what is running low
- Faster sales — cashiers scan barcodes and print receipts in seconds
- Accountability — every sale, stock change, and cash collection is tracked
- Better decisions — directors and managers get reports without manual spreadsheets
- Reduced losses — stock cannot be sold if it is not available; variances are flagged at cashier closing

2. Who Uses the System?

CSRMS supports three user roles. Each person sees only what they need — the Director cannot accidentally change stock, and Sales Agents cannot access management reports.

Role	Who they are	Main responsibilities
Director	Business owner / overseer	View company sales, inventory value, procurement totals, and performance reports. Download PDF reports.
Branch Manager	Day-to-day branch leader	Manage categories, products, prices, procurement, stock adjustments.
Sales Agent	Cashier / front-line staff	Search products, scan barcodes, process cash sales, print receipts, and view available stock.

3. Default Login Credentials (Demo)

When the system is first set up, three demo accounts are created for testing and training:

Role	Username	Password
Director	director	director123
Branch Manager	manager	manager123
Sales Agent	agent	agent123

Important: Change these passwords before going live in production.

This is the end-to-end process the system supports — from receiving goods to reviewing company performance:

1. Manager creates product categories (e.g. Refreshments, Groceries, Dairy)
2. Manager adds products with prices, barcodes, and reorder levels
3. Supplier delivers goods to the branch
4. Manager records procurement — stock increases automatically
5. Products become available for sale on the shop floor
6. Sales Agent searches or scans a product barcode at the till
7. System checks stock availability before allowing the sale
8. Customer pays cash — sale is recorded and receipt is printed
9. Stock is reduced automatically — no manual counting needed
10. Manager performs cashier balancing at end of day
11. Reports are updated in real time
12. Director reviews company performance and downloads PDF summaries

5. Module-by-Module Guide

Each section of the system maps directly to a part of your store operations:

5.1 Authentication & Security

- Staff log in with a username and password
- Passwords are encrypted — never stored in plain text
- Sessions expire after 30 minutes of inactivity
- Each user is assigned one role that controls what they can see and do
- Password reset is available without administrator intervention

5.2 Dashboard (Home Screen)

- Director sees: total company sales, inventory value, procurement value, top categories and products
- Manager sees: today's sales, inventory summary, low-stock alerts, cashier sales breakdown
- Sales Agent sees: daily sales total, quick access to new sale, recent transactions

5.3 Category Management

- Organises products into groups (Refreshments, Groceries, Dairy Products, etc.)
- Manager can create, edit, activate, or deactivate categories
- A category must exist before a product can be added — enforces proper organisation

5.4 Product Management

- Each product has: name, category, cost price, selling price, stock quantity, reorder level, and barcodes
- Managers set and update selling prices
- Multiple barcodes can be assigned to one product (useful for different pack sizes)

- Inactive products cannot be sold

5.5 Procurement (Stock Receiving)

- Records goods received from suppliers: product, supplier name, quantity, cost, and date
- When procurement is saved, stock increases automatically: $\text{Current Stock} + \text{Quantity Received}$
- Example: 100 units in stock + 50 received = 150 units updated instantly
- Every procurement is linked to the manager who recorded it

5.6 Inventory Management

- View all products with current stock levels and status (In Stock / Low Stock / Out of Stock)
- Stock increases automatically when procurement is recorded
- Stock decreases automatically when a sale is completed
- Managers can make manual adjustments (e.g. -10 for damaged goods) with a required reason
- Full history of every stock movement is kept for audit purposes

5.7 Sales (Point of Sale)

- Sales Agent scans a barcode or searches by product name
- System retrieves price and checks stock before allowing the sale
- Cash payment is entered; system calculates total and change
- Sale cannot complete if stock is insufficient or payment is too low
- Receipt is generated immediately with receipt number, date, product details, and agent name

5.8 Cashier Balancing

- At end of day, manager reviews each cashier's total sales
- Cashier submits the actual cash collected
- System compares sales total vs. cash collected and records any variance
- Manager approves the daily balance — creating accountability for cash handling

5.9 Receipts

- Every completed sale produces a receipt with: receipt number, date/time, product, quantity, price, total, agent, and branch
- Receipts can be printed directly from the browser
- PDF receipts can be downloaded for records or customer copies

5.10 Reports & Analytics

Report	Who can view	What it shows
Daily Sales	Manager, Director	Products sold, quantities, revenue for a selected day
Inventory	Manager, Director	Current stock, low-stock items, out-of-stock items, total value
Procurement	Manager, Director	Products received, suppliers, quantities, costs
Cashier Balancing	Manager	Per-cashier sales, cash collected, variances
Company Performance	Director only	Total sales, inventory value, procurement value, top sellers

All reports can be exported as PDF for meetings, filing, or sharing with stakeholders.

The system proactively warns managers before stock problems affect customers:

- Low Stock Alert — when quantity falls at or below the reorder level (e.g. 10 units left, reorder level is 20)
- Out of Stock Alert — when quantity reaches zero; product becomes unavailable for sale automatically
- Alerts appear on the Manager dashboard and in the Notifications section

7. Business Rules Enforced by the System

These rules are built into the software and cannot be bypassed by staff:

1. A category must exist before a product can be created
2. Every product must belong to a category
3. A product may have one or more barcodes
4. Only Managers can create categories, products, and update prices
5. Only products with available stock can be sold
6. Stock reduces automatically after every sale
7. Sales Agents cannot modify inventory or access management reports
8. All inventory movements are recorded in an audit log
9. Directors can only view consolidated company information — no operational changes

8. How the Technology Works (Simple Overview)

For stakeholders who want to understand what powers the system without technical jargon:

The three main components

- Web Application (Node.js + Express) — the brain of the system; handles login, business logic, calculations, and security
- Web Pages (EJS templates) — what staff see in the browser; dashboards, forms, tables, and receipts
- Database (PostgreSQL) — securely stores all products, sales, users, stock movements, and reports data on your laptop/server

How a sale works behind the scenes

1. Sales Agent submits the sale form in the browser
2. Server verifies the user is logged in and has Sales Agent role
3. Server checks product stock in the database (locks the record to prevent double-selling)
4. Server calculates total, records the sale, reduces stock, and logs the inventory movement
5. Server generates a unique receipt number and shows the receipt page
6. If stock drops below reorder level, a notification is created for the Manager

- Encrypted passwords (bcrypt hashing)
- Role-based access control on every page and action
- Session timeout after 30 minutes
- Audit log records logins, stock changes, sales, and approvals
- Database transactions ensure stock and sales stay in sync — no partial updates

The application follows a clear structure that mirrors your business modules:

Folder / Area	Purpose
server.js	Starts the application and connects all modules
routes/	Handles web requests — one file per module (sales, inventory, reports, etc.)
views/	HTML page templates staff see in the browser
database/schema.sql	Database structure — tables for products, sales, users, etc.
middleware/auth.js	Security — login checks and role permissions
public/css/	Visual design and layout styling

10. Presentation Demo Script

Suggested order for demonstrating CSRMS to store owners (approx. 15–20 minutes):

1. Log in as Director — show the high-level dashboard and Company Performance report
2. Log out, log in as Manager — create a category and add a new product with barcode
3. Record a procurement — show stock increase on the inventory screen
4. Log in as Sales Agent — scan/search product, complete a cash sale, print receipt
5. Return as Manager — show stock decreased, view inventory history, check low-stock alert
6. Demonstrate cashier balancing — submit cash, show variance calculation, approve balance
7. End as Director — download PDF performance report for the meeting

11. System Requirements

- Computer or server running Windows, Mac, or Linux
- Node.js 18 or higher installed
- PostgreSQL database (installed on your laptop or a dedicated server)
- Modern web browser (Chrome, Edge, or Firefox)
- Local network or internet access for staff devices
- Supports 500+ transactions per day with response times under 3 seconds

12. Getting Started (Quick Reference)

1. Install Node.js and PostgreSQL on the host machine
2. Extract the project and run: `npm install`
3. Copy `.env.example` to `.env` and set the database password
4. Run: `npm run db:init` (creates database, tables, and sample data)
5. Run: `npm start` and open `http://localhost:3000`

6. Train staff using the demo accounts, then create real user accounts
7. Replace demo passwords before production use

General Workflow July 2026

— End of Document —